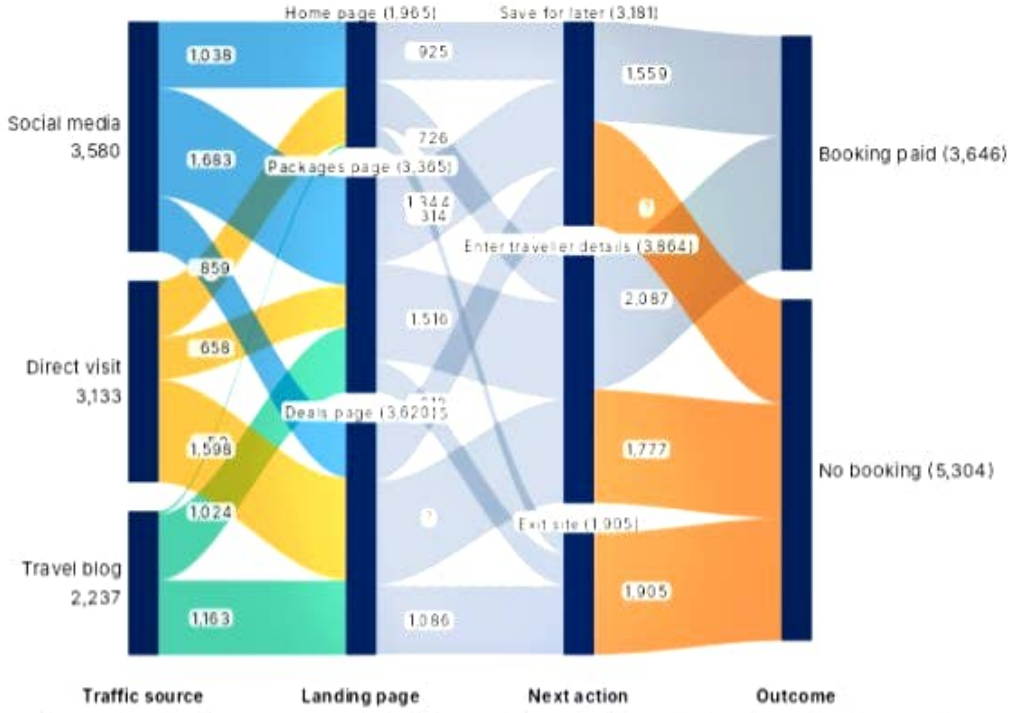
R.No: 4 Reg. No: 192424243 Name: B PALLAVI

Scenario: The analytics team of a travel-booking website tracked 8,950 sessions last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of sessions on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Deals page to Enter traveller details; Direct visit to Home page; Save for later to No booking. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Booking paid as a percentage of all sessions; (ii) the largest single flow into "No booking" or "Exit site"; (iii) the percentage of "Deals page" sessions that go on to "Enter traveller details". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 400 sessions. Compute the height in cm of each landing-page node (Deals page, Packages page, Home page). [2]

(e) Write R code: a links data frame for the first four flows leaving "Direct visit" and "Social media" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

Activity-6.

(a) Hidden flows.

Rule : Inflow = outflow.

1. Deals → Enter traveller details : 3620 - 2015 = 1598 = 7 sessions.
2. Direct visit → Home Page : 3133 - 859 - 658 - 1598 = 18 sessions
3. save for later → No booking : 3181 - 1559 = 1622 sessions.

(b) Calculations.

i, conversion rate

$$\frac{3646}{8950} \times 100 = 40.74\%.$$

ii, Largest flow through exit site = 1405 sessions.

iii, Deals → Enter traveller details.

$$\frac{519}{3620} \times 100 = 14.34\%.$$

(c) Design decisions.

1. Node order : Arrange stages from traffic source to outcome.
2. Colour : Use different colours for sources & outcomes
3. Labels : show node names & session counts.
4. Highlight : Emphasize Paid bookings & major drop-off

(d) Node heights.

Height = sessions / 400.

Deals Page : 3620 / 400 = 9.05 cm

Packages : 3365/400 = 8.41 cm

Home : 1965/400 = 4.01 cm

(e) R code

```
library (networks)
links <- data, frame (
    source = c(0,0,1,1),
    target = c(3,4,3,5),
    Value = c(1038, 1683, 859, 1598).
)
nodes <- data.frame (.
    name = c("social media", "Direct Visit",
             "Travel blog", "Home Page",
             "Packages page", "Deals page")
    ).
sankey Network (.
        Links = links,
        Nodes = nodes,
        Source = "source",
        Target = "target",
        Value = "Value",
        NodeID = "name".
    ).
```

(f) Limitations and alternatives.

* Too many crossing flows make the diagram confusing use a bar chart.
* Small flows are difficult to compare, use a table with exact values.